

Supply Chain Risk & Financial Analysis Documentation

1. Project Overview

The objective of this project was to identify financial vulnerabilities and geographic concentration risks within a corporate supply chain. By integrating procurement commitments with actual payment data, the project provides a 360-degree view of vendor reliability and liquidity.

2. Data Acquisition & Collection

Method: Data was generated using Google Gemini via Python integration.

Environment: Google Colab.

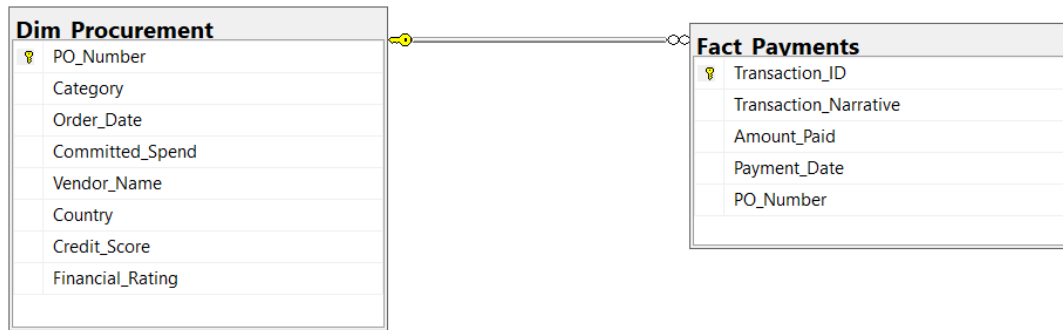
Source Data: Three primary datasets were created: Procurement_data.csv, Bank_ledger.csv, and Vendor_risk.csv.

3. Data Cleaning Process (Python/Colab)

- The cleaning process in the Data_Cleaning_Process.ipynb notebook involved the following steps:
- Library Imports: Utilized pandas for data manipulation and numpy for handling missing values.
- Standardization: Converted Order_Date and Payment_Date to a uniform datetime format for accurate time-series analysis.
- Data Validation: Checked for duplicate PO_Number entries to maintain referential integrity.
- Currency Normalization: Cleaned financial columns (Committed_Spend, Amount_Paid) by removing non-numeric characters and ensuring they were stored as floats.
- Merging: Combined vendor risk ratings with procurement data to create a unified dimension table.

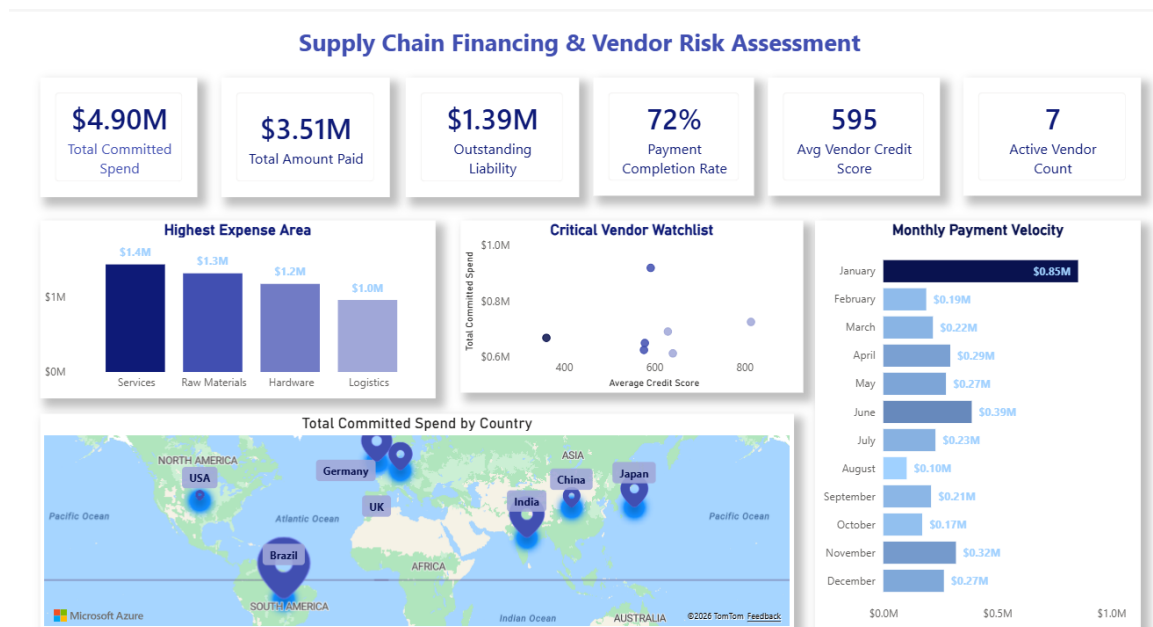
4. Data Modeling (SQL)

- A Star Schema was developed in SQL Server to optimize the data for Power BI performance:
- Dimension Table (Dim_Procurement): Contains descriptive attributes such as Category, Vendor_Name, Country, and Credit_Score.
- Fact Table (Fact_Payments): Contains quantitative transaction data, including Amount_Paid and Payment_Date.
- Relationship: A 1-to-Many relationship was established via the PO_Number key.



5. Data Visualization & Business Solutions

- The Power BI dashboard addresses the following critical business problems:
- Spend Visibility: A Clustered Bar Chart identifies that Logistics and Marketing are the highest expense areas, highlighting where negotiation leverage is strongest.
- Financial Risk: A Scatter Plot uses conditional formatting to flag vendors with low credit scores and high spend (e.g., Vendor_9), allowing for proactive backup planning.
- Geographic Exposure: A Map Visual reveals concentration in Canada and the UK, prompting a strategy for regional diversification.
- Liquidity Management: KPI cards for Outstanding Liability (\$56,194.05) allow the finance team to forecast short-term debt accurately.



6. Conclusion

This project successfully moved the procurement process from reactive to proactive. By visualizing the "Gap" between commitments and payments, the organization can now manage its cash flow more effectively while significantly reducing its exposure to unstable suppliers.